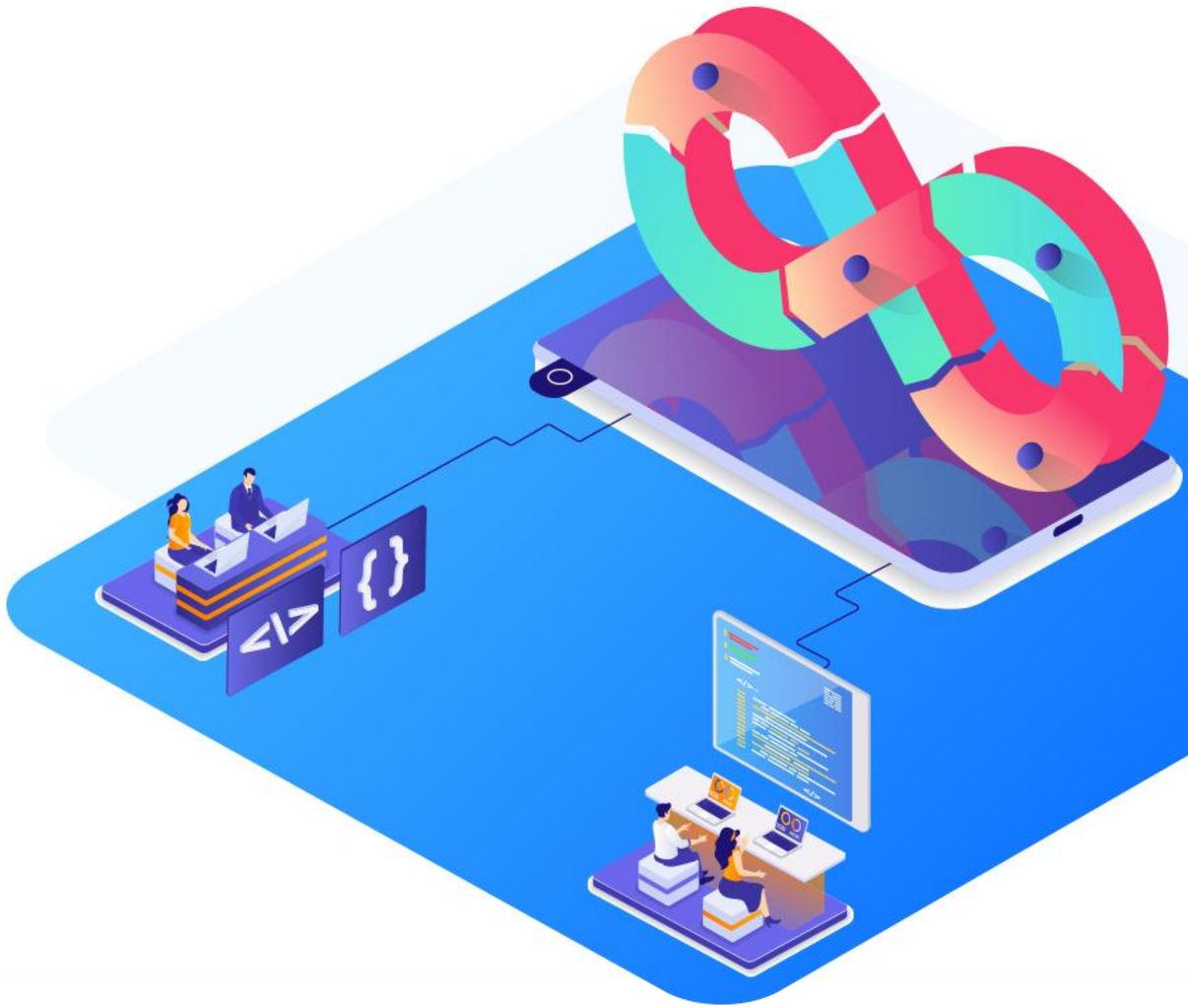
Container Orchestration Using Kubernetes



Course-End Project

Deploy an Application Using the Kubernetes Dashboard

Objective

To deploy a multi-tier MySQL application using Kubernetes with specific configurations for user roles, storage, service verification, namespace restrictions, quota limits, and data management

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Problem Statement and Motivation

Real-time scenario:

Karen is a DevOps engineer at a tech startup. Her team has developed a new application using MySQL. Now, it is her task to deploy that application.

The company plans to utilize Kubernetes for its robust container orchestration capabilities.

Karen must create a Kubernetes dashboard with specific configurations, user roles, storage, service verification, and data management.

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Industry Relevance

The following tools used in this project serve specific purposes within the industry:

1. **kubeadm:** A utility that offers **kubeadm init** and **kubeadm** join as efficient ways to bootstrap Kubernetes clusters. It focuses on bootstrapping rather than machine provisioning.

2. **kubectl:** A command-line interface for Kubernetes that allows execution of commands against Kubernetes clusters. It can be used for deploying applications, managing cluster resources, and viewing logs.

3. **kubelet:** An essential node agent present on every node in a Kubernetes cluster. It ensures that the containers described in the provided PodSpecs are running and healthy.

4. **Docker:** Docker is a tool designed to facilitate developers in building, sharing, and running applications in containers. It takes care of the setup, allowing developers to concentrate on the code.

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Tasks

The following tasks outline the process of deploying an application using Kubernetes:

1. Get started with pods, services, and deployments
2. Create and verify the service
3. Create a token and work on a dashboard
4. Configure the NFS-server for MySQL and WordPress deployment
5. Set up the NFS client side
6. Create and verify the PV
7. Create a secret for MySQL deployments secret data
8. Create a configmap for WordPress deployment to store non-sensitive information

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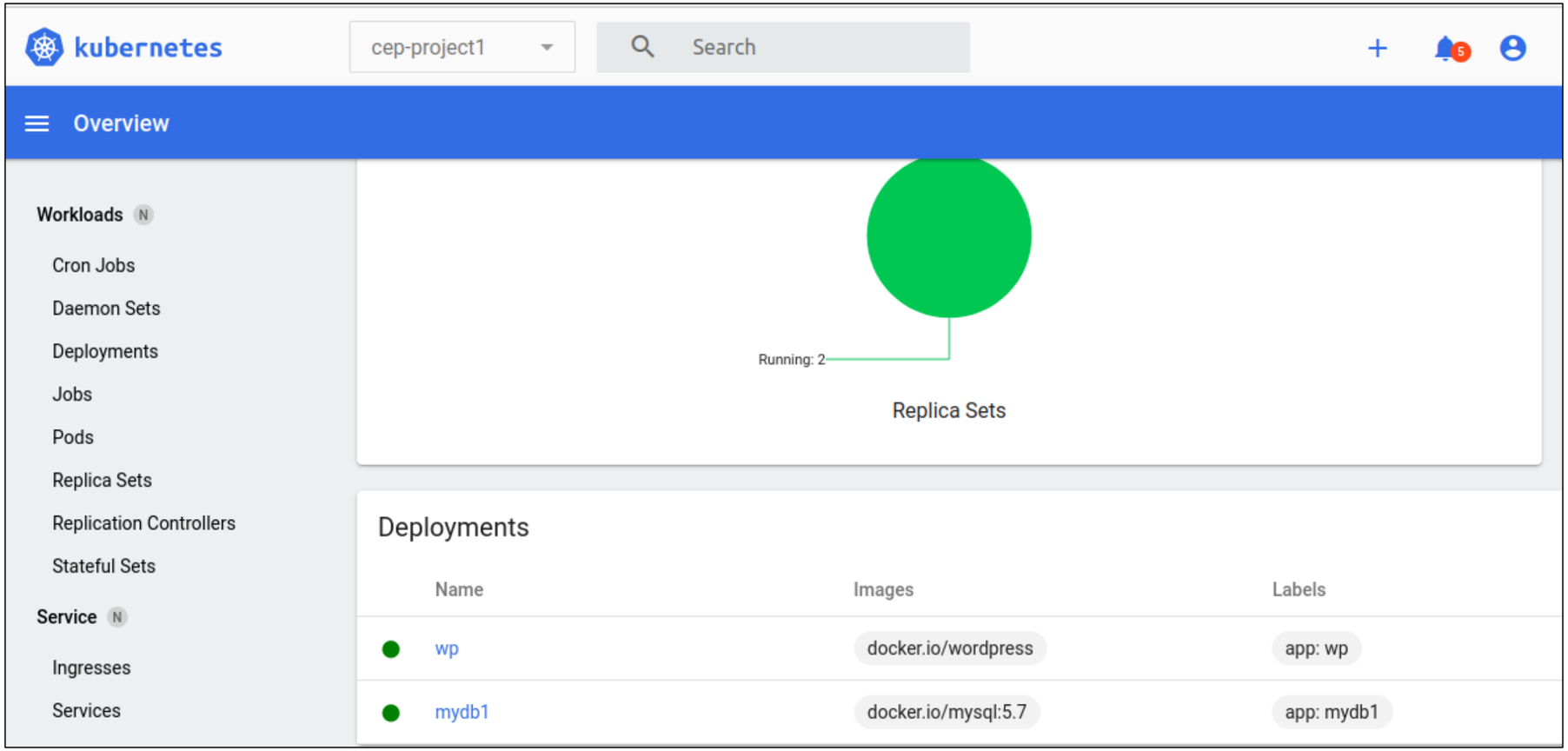
Project References

Task 1: Lesson 2
Task 2: Lesson 6
Task 3: Lesson 3
Task 4: Lesson 4
Task 5: Lesson 4
Task 6: Lesson 6
Task 7: Lesson 4
Task 8: Lesson 4

Output Screenshots



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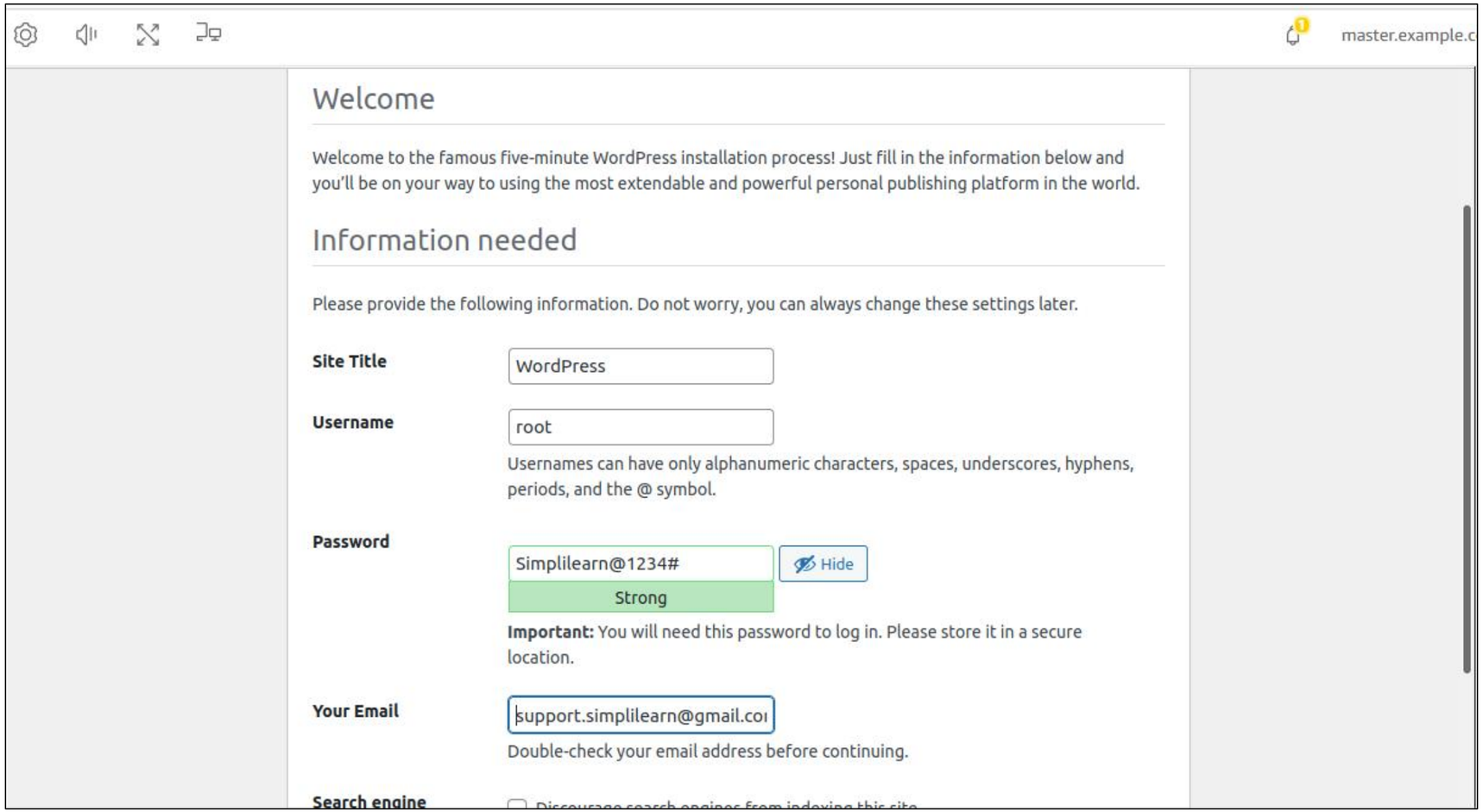

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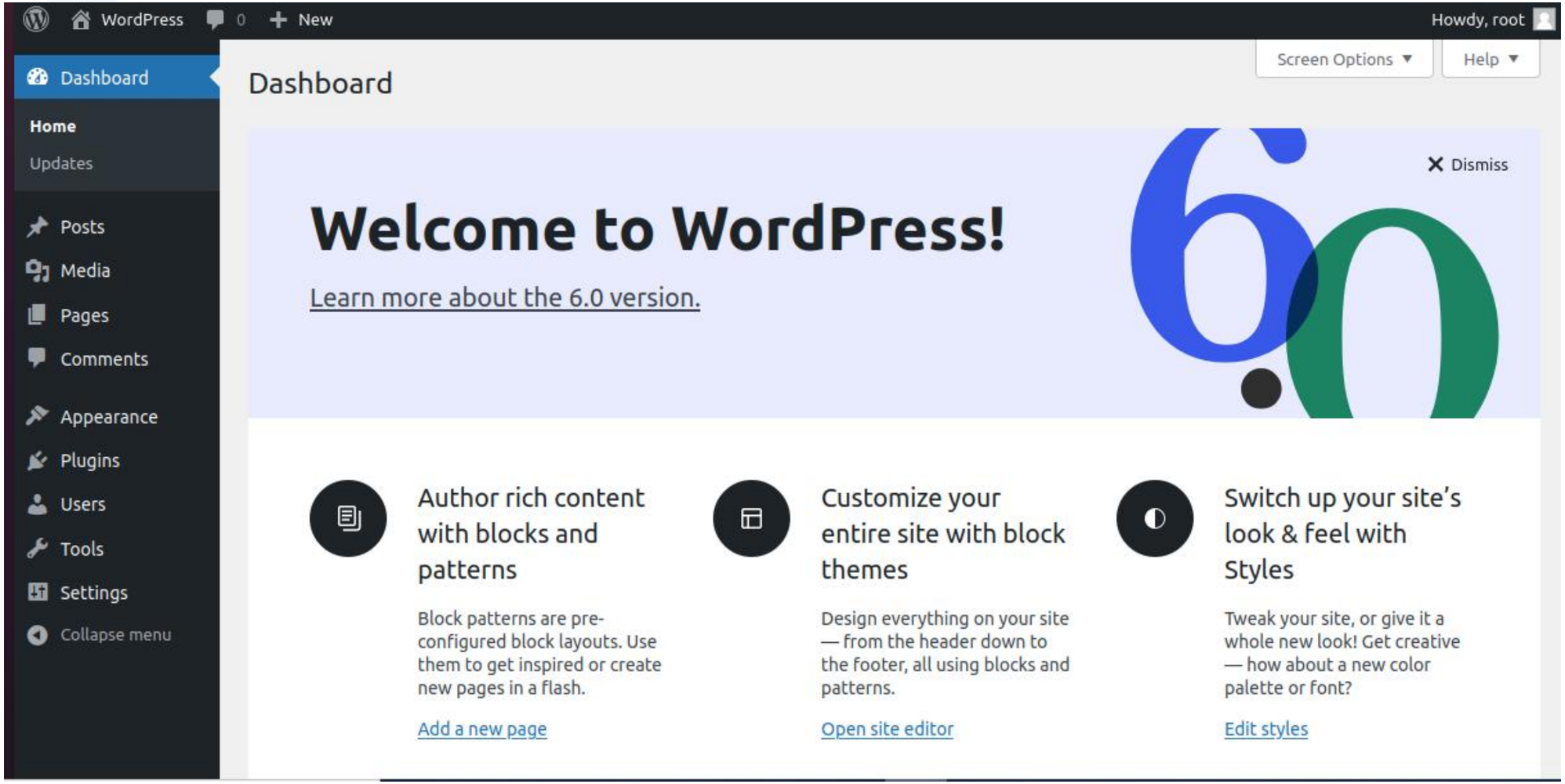

Thank you